

Mecklenburg County Sample — Shopping Center / Retail



Traffic Impact Study

Prepared August 11, 2026 · 35.1954, -80.7643

Simple Impact Studies — sample draft

Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Mecklenburg County Sample — Shopping Center / Retail within Charlotte MSA, North Carolina (City of Charlotte / Mecklenburg County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 15 study intersections within a 1.00-mile radius are analyzed under Existing, No-Build, and Build conditions.

15	0	11	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Mecklenburg County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.1954°, -80.7643°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Charlotte / Mecklenburg County
MPO	CRTPO (Charlotte Regional Transportation Planning Organization) — 2026-2035 TIP
Programmed projects source	NCDOT 2026-2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Charlotte UDO Article 32, Section 32.1 (effective 6/1/2023) requires a Comprehensive Transportation Review (CTR) — Multimodal Assessment + TDM Assessment + Traffic Impact Study — when Charlotte Streets Manual Chapter 3, Table 3.1 thresholds are met (reported secondary-source values: >1,500 vpd or >150 peak-hour trips in lower-intensity districts; >2,000 vpd or >200 peak-hour trips in medium/high-intensity districts — confirm against the current Table 3.1 at scoping). The CTR applies at permitting even for by-right development.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

CDOT publishes a Rezoning Transportation Analysis memo per rezoning petition; the TIS must be prepared by a North Carolina licensed Professional Engineer. NCDOT coordination applies on state-system streets. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening analysis applies HCM-consistent computation; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	North Sharon Amity Road & Holbrook Drive	B / 14.6s	B / 14.6s	B / 17.1s
2	Near North Sharon Amity Road	F / 120.0s	F / 120.0s	F / 120.0s
3	North Sharon Amity Road & Charleston Drive	B / 16.3s	B / 16.3s	B / 17.6s
4	Albemarle Road & North Sharon Amity Road	F / 120.0s	F / 120.0s	F / 120.0s
5	Albemarle Road & Pierson Drive	B / 14.1s	B / 14.1s	B / 14.2s
6	North Sharon Amity Road & Monroe Road	F / 120.0s	F / 120.0s	F / 120.0s
7	Ashmore Dr & Monroe Rd	F / 116.1s	F / 120.0s	F / 120.0s
8	Idlewild Rd & Monroe Rd_rama Rd	F / 96.4s	F / 103.0s	F / 104.7s
9	Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	F / 120.0s	F / 120.0s	F / 120.0s

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
10	Idlewild Road & East Independence Expressway	E / 70.1s	E / 75.2s	E / 76.3s
11	Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	F / 120.0s	F / 120.0s	F / 120.0s
12	Mcclintock Middle School	F / 116.1s	F / 120.0s	F / 120.0s
13	Flasher Sheffield Dr & Woodland Dr	F / 120.0s	F / 120.0s	F / 120.0s
14	Oakhurst Elementary School	B / 14.4s	B / 14.4s	B / 14.4s
15	Albemarle Rd & Reddman Rd	F / 120.0s	F / 120.0s	F / 120.0s

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	North Sharon Amity Road & Holbrook Drive	+17%	no	no	none indicated
2	Near North Sharon Amity Road	+0%	no	YES	MITIGATION INDICATED
3	North Sharon Amity Road & Charleston Drive	+8%	no	no	none indicated
4	Albemarle Road & North Sharon Amity Road	+0%	no	YES	MITIGATION INDICATED
5	Albemarle Road & Pierson Drive	+1%	no	no	none indicated
6	North Sharon Amity Road & Monroe Road	+0%	no	YES	MITIGATION INDICATED
7	Ashmore Dr & Monroe Rd	+0%	no	YES	MITIGATION INDICATED
8	Idlewild Rd & Monroe Rd_rama Rd	+2%	no	YES	MITIGATION INDICATED
9	Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	+0%	no	YES	MITIGATION INDICATED
10	Idlewild Road & East Independence Expressway	+1%	no	no	none indicated
11	Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	+0%	no	YES	MITIGATION INDICATED
12	Mcclintock Middle School	+0%	no	YES	MITIGATION INDICATED
13	Flasher Sheffield Dr & Woodland Dr	+0%	no	YES	MITIGATION INDICATED
14	Oakhurst Elementary School	+0%	no	no	none indicated
15	Albemarle Rd & Reddman Rd	+0%	no	YES	MITIGATION INDICATED

5.0 CONCLUSIONS AND SUBMITTAL NOTES

0 intersections project a LOS drop and 11 operate at LOS E or F under Build conditions; mitigation per Policy Ch. 5.J should be developed with the District Engineer.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis

Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: $a=28507$, $b=-0.02$, $c=-0.123$) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Sharon Amity Road & Holbrook Drive	0.02	0.1	402	42.9%
Near North Sharon Amity Road	0.14	0.3	218	23.3%
North Sharon Amity Road & Charleston Drive	0.19	0.5	172	18.4%
Albemarle Road & North Sharon Amity Road	0.47	1.1	41	4.4%
Albemarle Road & Pierson Drive	0.58	1.4	24	2.6%
North Sharon Amity Road & Monroe Road	0.61	1.5	20	2.1%
Ashmore Dr & Monroe Rd	0.64	1.5	18	1.9%
Idlewild Rd & Monroe Rd_rama Rd	0.80	1.9	8	0.9%
Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block	0.83	2.0	7	0.7%
Idlewild Road & East Independence Expressway	0.85	2.0	6	0.6%
Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)	0.86	2.1	6	0.6%
Mcclintock Middle School	0.90	2.2	5	0.5%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 92% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 8% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

A capacity-constrained assignment using the BPR volume-delay function $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives until the assignment is congestion-consistent. Highest post-build v/c among study signals: 6.35. Per-signal v/c , delay, LOS, and queue are in the capacity appendix. (Road-network corridor routing was not available for this region; per-signal capacity-constrained assignment was used.)

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

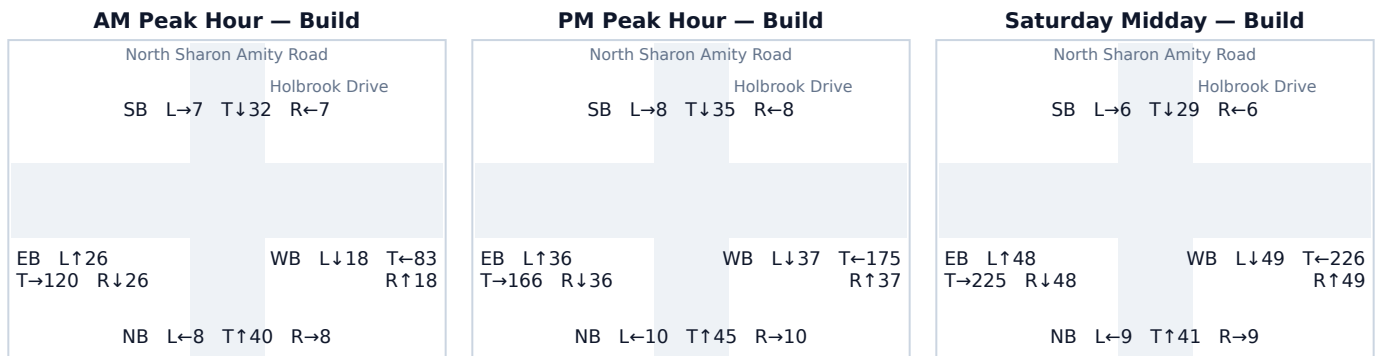
One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 North Sharon Amity Road & Holbrook Drive

Signal ID	charlotte-172660814
Location	NPA 16 · 0.02 mi from site
Added PM peak trips	402
Existing / No-Build (PM)	LOS B · 14.6 s/veh · v/c 0.11
Build (PM)	LOS B · 17.1 s/veh · v/c 0.33
Δ control delay	+2.5 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	55	10	65	0.07 → 0.08	14.2 → 14.3	B → B	38
SB	51	0	51	0.06 → 0.06	14.2 → 14.2	B → B	30
EB	55	183	238	0.07 → 0.29	14.2 → 16.6	B → B	156
WB	40	209	249	0.05 → 0.31	14.0 → 16.8	B → B	164

Affected movements (PM peak project trips)

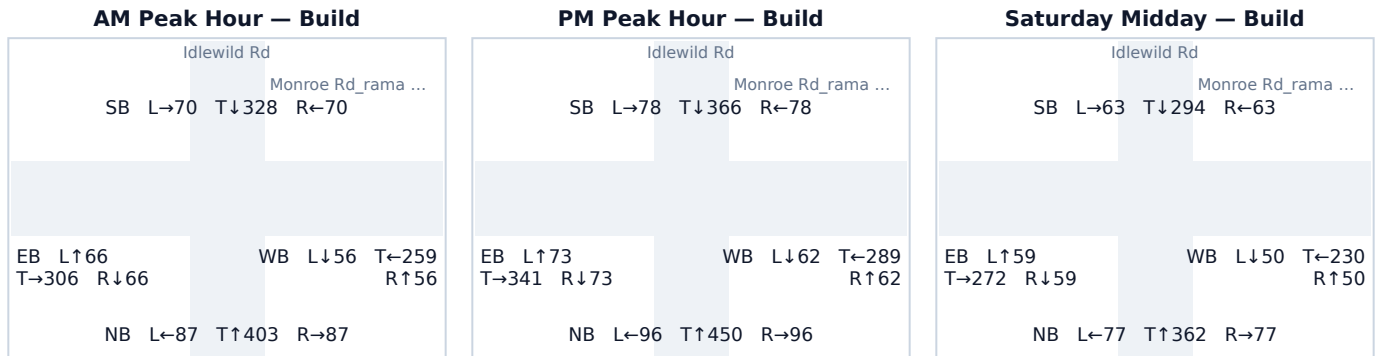
Movement	Project trips	% of project trips
WB Through	199	50%
EB Through	183	46%
WB Left	10	2%
NB Right	10	2%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 Idlewild Rd & Monroe Rd_rama Rd

Signal ID	charlotte-7945928189
Location	NPA 52 · 0.80 mi from site
Added PM peak trips	8
Existing / No-Build (PM)	LOS F · 103.0 s/veh · v/c 1.14
Build (PM)	LOS F · 104.7 s/veh · v/c 1.15
Δ control delay	+1.7 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	638	4	642	0.79 → 0.79	28.7 → 29.0	C → C	566
SB	518	4	522	0.64 → 0.64	23.0 → 23.1	C → C	417
EB	487	0	487	0.60 → 0.60	21.9 → 22.0	C → C	379
WB	413	0	413	0.51 → 0.51	19.9 → 19.9	B → B	304

Affected movements (PM peak project trips)

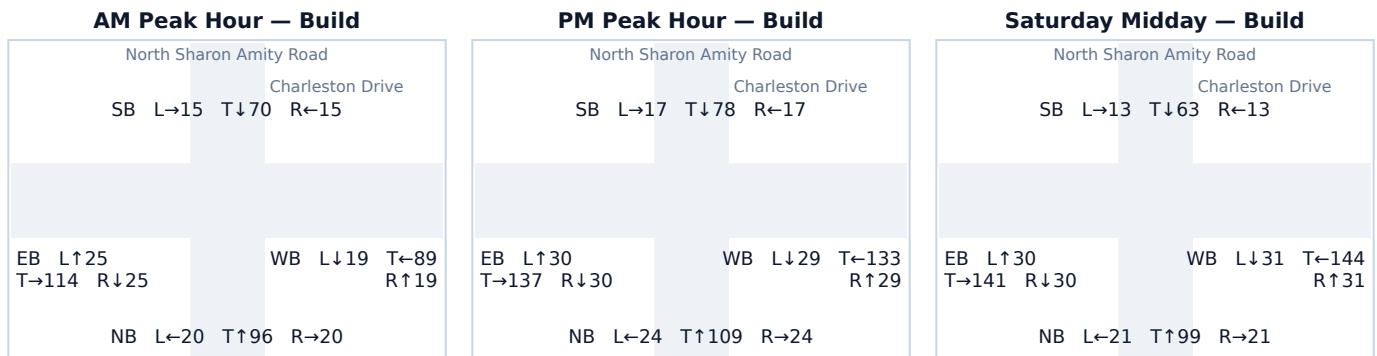
Movement	Project trips	% of project trips
SB Through	4	50%
NB Through	4	50%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 North Sharon Amity Road & Charleston Drive

Signal ID	charlotte-7810838990
Location	NPA 52 · 0.19 mi from site
Added PM peak trips	172
Existing / No-Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Build (PM)	LOS B · 17.6 s/veh · v/c 0.36
Δ control delay	+1.3 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	134	23	157	0.17 → 0.19	15.2 → 15.4	B → B	97
SB	112	0	112	0.14 → 0.14	14.9 → 14.9	B → B	68
EB	137	60	197	0.17 → 0.24	15.2 → 16.0	B → B	126
WB	101	89	191	0.13 → 0.24	14.7 → 15.9	B → B	121

Affected movements (PM peak project trips)

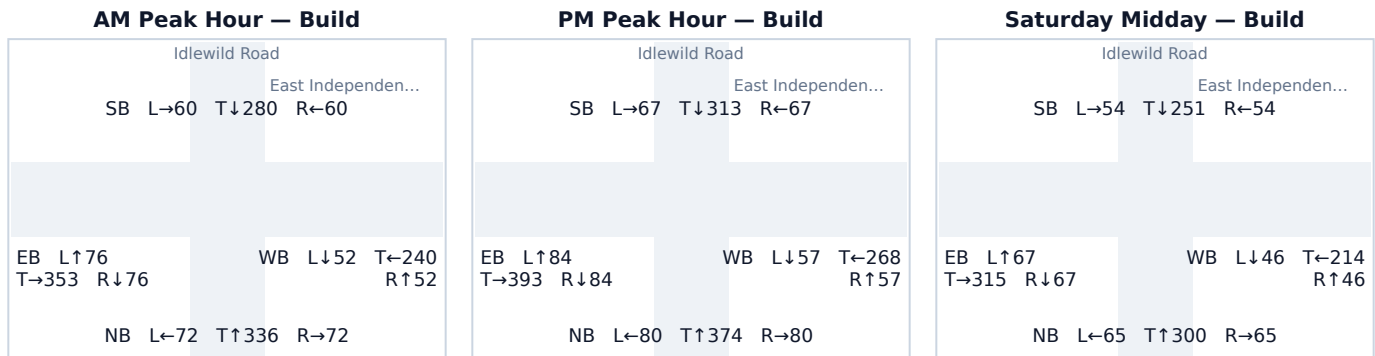
Movement	Project trips	% of project trips
WB Through	65	38%
EB Through	60	35%
WB Left	24	14%
NB Right	23	13%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Idlewild Road & East Independence Expressway

Signal ID	charlotte-5516526369
Location	NPA 52 · 0.85 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS E · 75.2 s/veh · v/c 1.07
Build (PM)	LOS E · 76.3 s/veh · v/c 1.07
Δ control delay	+1.1 s/veh
Mitigation	Moderate: extend critical-phase green time and consider a protected-only left-turn phase to absorb the new demand without queue spillback.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	531	3	534	0.66 → 0.66	23.4 → 23.5	C → C	431
SB	444	3	447	0.55 → 0.55	20.7 → 20.8	C → C	337
EB	561	0	561	0.69 → 0.69	24.6 → 24.6	C → C	463
WB	382	0	382	0.47 → 0.47	19.3 → 19.3	B → B	275

Affected movements (PM peak project trips)

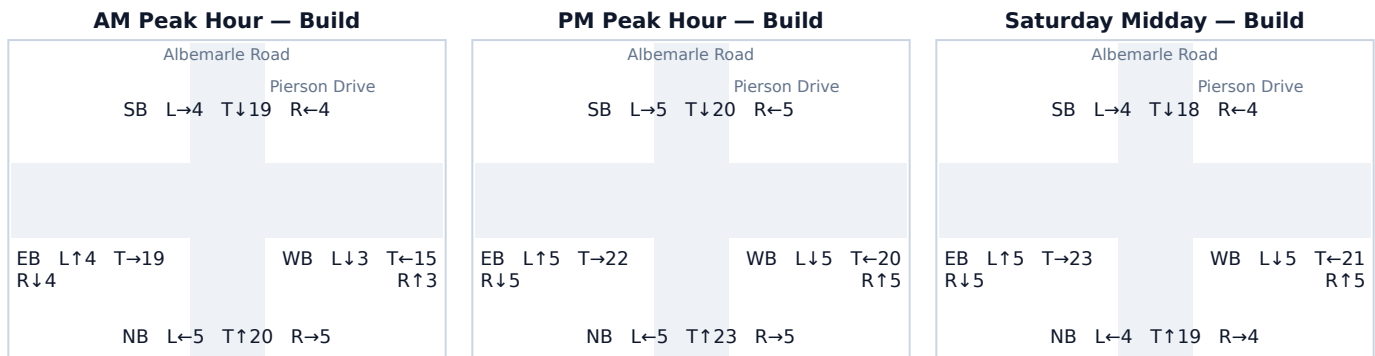
Movement	Project trips	% of project trips
SB Through	3	50%
NB Through	3	50%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Albemarle Road & Pierson Drive

Signal ID	charlotte-1711278797
Location	NPA 240 · 0.58 mi from site
Added PM peak trips	24
Existing / No-Build (PM)	LOS B · 14.1 s/veh · v/c 0.06
Build (PM)	LOS B · 14.2 s/veh · v/c 0.07
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	33	0	33	0.04 → 0.04	14.0 → 14.0	B → B	19
SB	28	2	30	0.03 → 0.04	13.9 → 13.9	B → B	18
EB	23	10	32	0.03 → 0.04	13.9 → 14.0	B → B	19
WB	17	12	30	0.02 → 0.04	13.8 → 13.9	B → B	17

Affected movements (PM peak project trips)

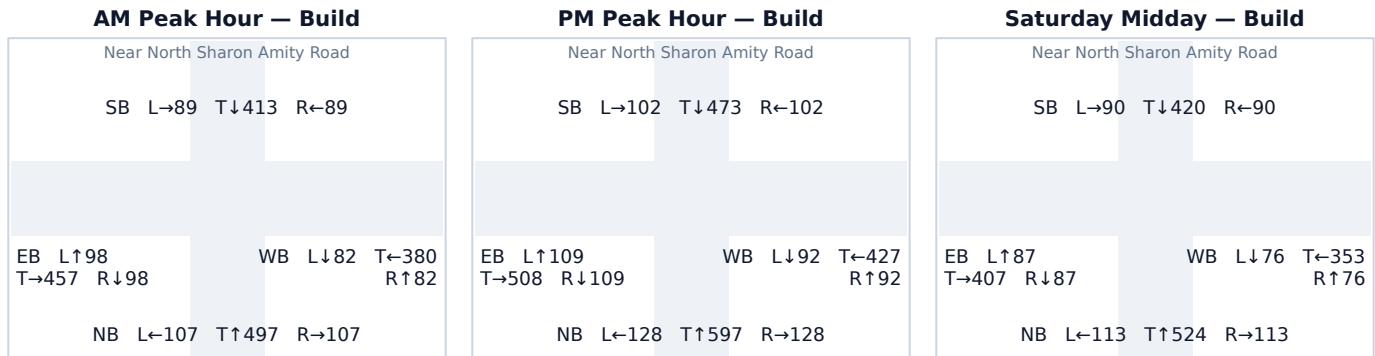
Movement	Project trips	% of project trips
WB Through	10	42%
EB Through	10	42%
WB Right	2	8%
SB Left	2	8%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Near North Sharon Amity Road

Signal ID	charlotte-2181076118
Location	NPA 16 · 0.14 mi from site
Added PM peak trips	218
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.47
Build (PM)	LOS F · 120.0 s/veh · v/c 1.59
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	740	113	853	0.91 → 1.05	39.6 → 71.2	D → E	873
SB	595	82	677	0.74 → 0.84	26.2 → 31.8	C → C	616
EB	726	0	726	0.90 → 0.90	37.4 → 37.4	D → D	690
WB	588	23	611	0.73 → 0.75	25.8 → 27.0	C → C	524

Affected movements (PM peak project trips)

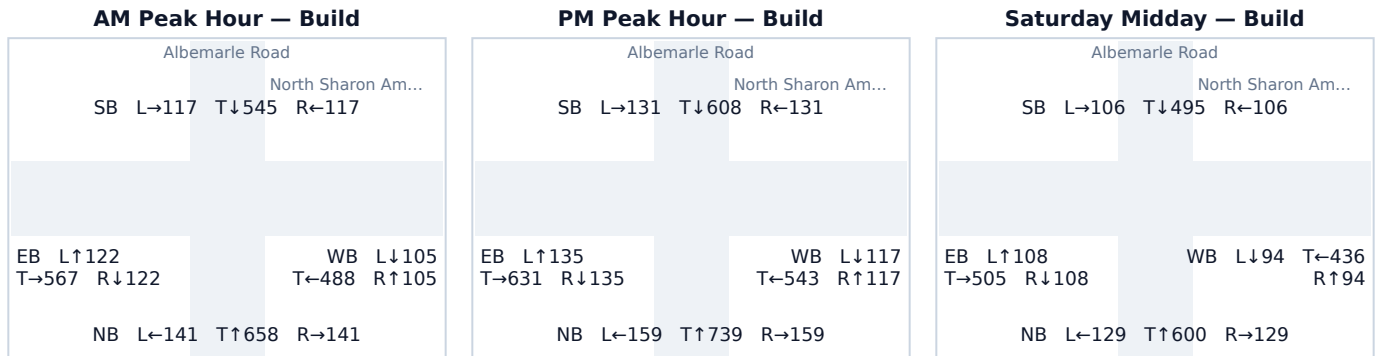
Movement	Project trips	% of project trips
NB Through	89	41%
SB Through	82	38%
NB Right	24	11%
WB Left	23	11%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Albemarle Road & North Sharon Amity Road

Signal ID	charlotte-2181076171
Location	NPA 16 · 0.47 mi from site
Added PM peak trips	41
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.98
Build (PM)	LOS F · 120.0 s/veh · v/c 2.00
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,035	21	1,057	1.28 → 1.30	120.0 → 120.0	F → F	1,081
SB	854	16	870	1.05 → 1.07	71.3 → 77.8	E → E	890
EB	901	0	901	1.11 → 1.11	91.6 → 91.6	F → F	922
WB	773	4	777	0.95 → 0.96	46.2 → 47.1	D → D	775

Affected movements (PM peak project trips)

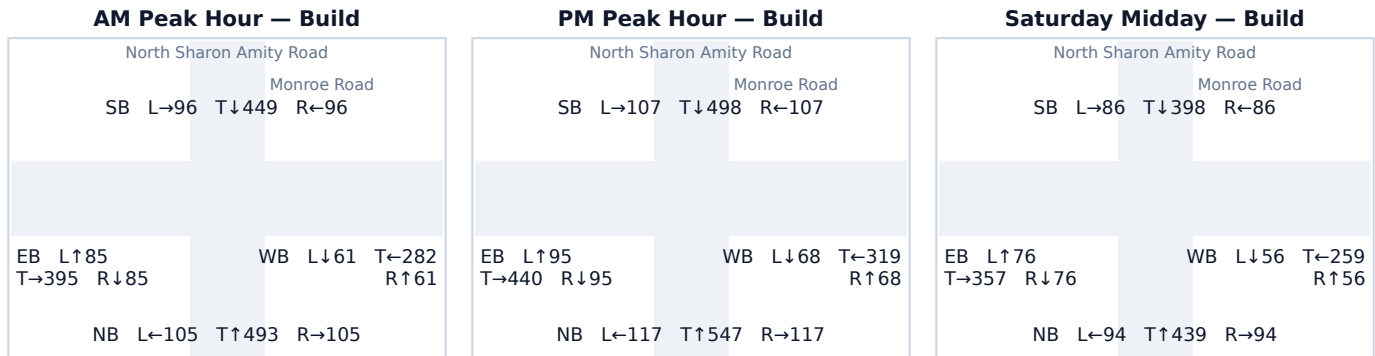
Movement	Project trips	% of project trips
NB Through	17	41%
SB Through	16	39%
NB Right	4	10%
WB Left	4	10%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 North Sharon Amity Road & Monroe Road

Signal ID	charlotte-5527699422
Location	NPA 52 · 0.61 mi from site
Added PM peak trips	20
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.42
Build (PM)	LOS F · 120.0 s/veh · v/c 1.43
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	778	3	781	0.96 → 0.96	47.4 → 48.2	D → D	783
SB	712	0	712	0.88 → 0.88	35.5 → 35.5	D → D	668
EB	623	6	630	0.77 → 0.78	27.8 → 28.2	C → C	549
WB	445	11	455	0.55 → 0.56	20.7 → 21.0	C → C	346

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	7	35%
EB Through	6	30%
WB Left	4	20%
NB Right	3	15%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 Ashmore Dr & Monroe Rd

Signal ID

charlotte-cdot-1896

Location

NPA 52 · 0.64 mi from site

Added PM peak trips

18

Existing / No-Build (PM)

LOS F · 120.0 s/veh · v/c 1.19

Build (PM)

LOS F · 120.0 s/veh · v/c 1.20

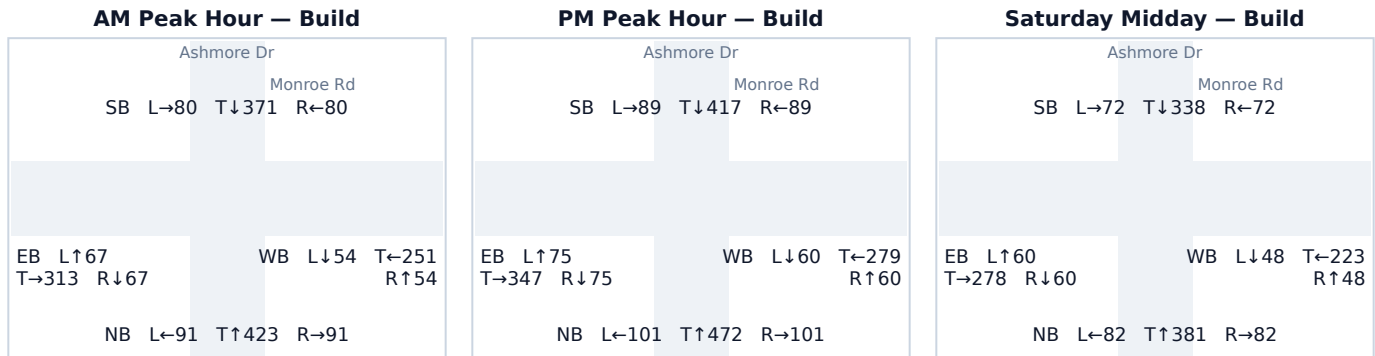
Δ control delay

+0.0 s/veh

Mitigation

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	667	7	674	0.82 → 0.83	30.9 → 31.5	C → C	612
SB	586	10	595	0.72 → 0.73	25.7 → 26.2	C → C	504
EB	495	1	497	0.61 → 0.61	22.2 → 22.2	C → C	389
WB	399	0	399	0.49 → 0.49	19.6 → 19.6	B → B	291

Affected movements (PM peak project trips)

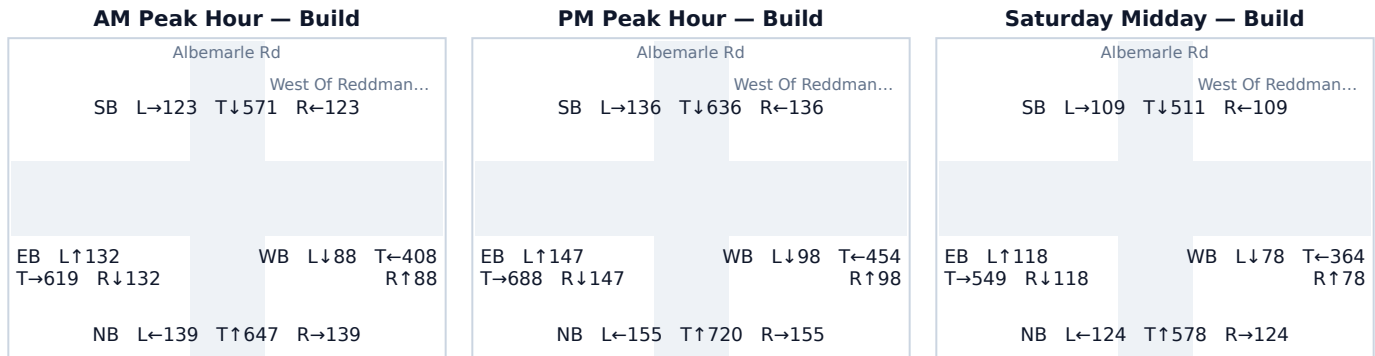
Movement	Project trips	% of project trips
SB Through	8	44%
NB Through	7	39%
SB Right	2	11%
EB Left	1	6%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Albemarle Rd & West Of Reddman Rd Ped Beacon Mid Block

Signal ID	charlotte-cdot-1721
Location	NPA 164 · 0.83 mi from site
Added PM peak trips	7
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.98
Build (PM)	LOS F · 120.0 s/veh · v/c 1.98
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,027	4	1,030	1.27 → 1.27	120.0 → 120.0	F → F	1,054
SB	906	2	908	1.12 → 1.12	93.9 → 95.1	F → F	929
EB	982	0	982	1.21 → 1.21	120.0 → 120.0	F → F	1,004
WB	649	1	650	0.80 → 0.80	29.5 → 29.5	C → C	576

Affected movements (PM peak project trips)

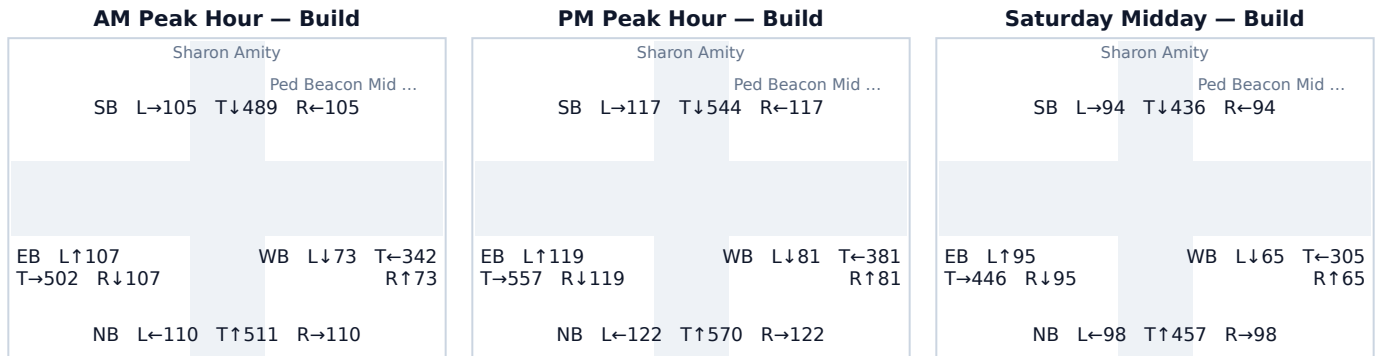
Movement	Project trips	% of project trips
NB Through	3	43%
SB Through	2	29%
NB Right	1	14%
WB Left	1	14%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Sharon Amity & Ped Beacon Mid Block (between Central Av & Albemarle Rd)

Signal ID	charlotte-cdot-1614
Location	NPA 164 · 0.86 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 1.62
Build (PM)	LOS F · 120.0 s/veh · v/c 1.63
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	811	4	814	1.00 → 1.00	56.3 → 57.3	E → E	832
SB	775	2	778	0.96 → 0.96	46.7 → 47.3	D → D	777
EB	795	0	795	0.98 → 0.98	51.9 → 51.9	D → D	808
WB	542	0	543	0.67 → 0.67	23.9 → 23.9	C → C	441

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	3	50%
SB Through	2	33%
NB Right	1	17%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 McClintock Middle School

Signal ID

charlotte-cdot-80017

Location

NPA 99 · 0.90 mi from site

Added PM peak trips

5

Existing / No-Build (PM)

LOS F · 120.0 s/veh · v/c 1.19

Build (PM)

LOS F · 120.0 s/veh · v/c 1.20

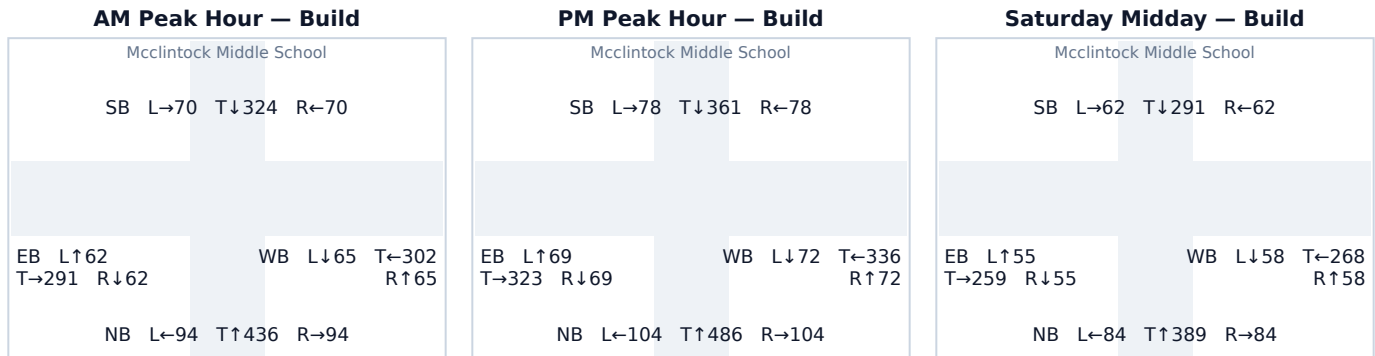
Δ control delay

+0.0 s/veh

Mitigation

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	692	2	694	0.85 → 0.86	33.3 → 33.5	C → C	641
SB	514	3	517	0.63 → 0.64	22.8 → 22.9	C → C	411
EB	461	0	461	0.57 → 0.57	21.2 → 21.2	C → C	352
WB	480	0	480	0.59 → 0.59	21.7 → 21.7	C → C	371

Affected movements (PM peak project trips)

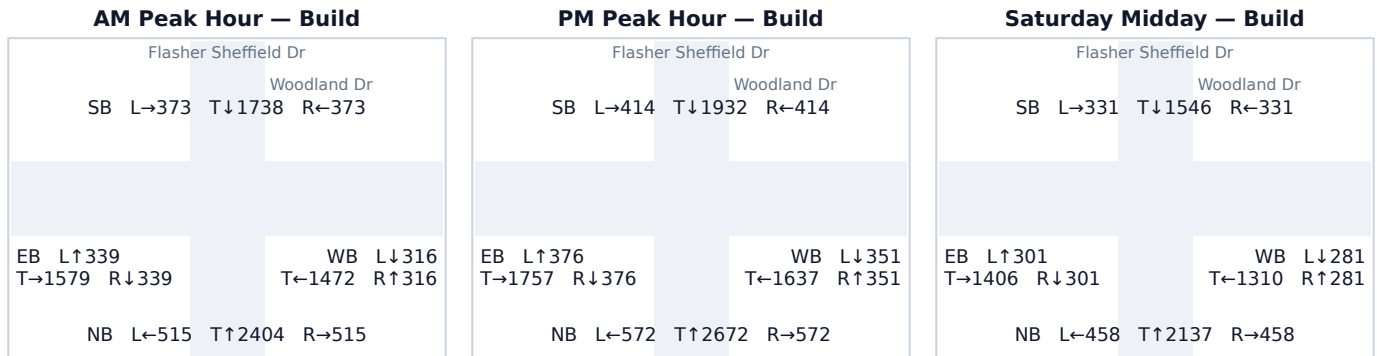
Movement	Project trips	% of project trips
SB Through	2	40%
NB Through	2	40%
SB Right	1	20%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Flasher Sheffield Dr & Woodland Dr

Signal ID	charlotte-172295413
Location	NPA 365 · 0.93 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS F · 120.0 s/veh · v/c 6.34
Build (PM)	LOS F · 120.0 s/veh · v/c 6.35
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	3,816	0	3,816	4.71 → 4.71	120.0 → 120.0	F → F	3,903
SB	2,760	0	2,760	3.41 → 3.41	120.0 → 120.0	F → F	2,823
EB	2,507	1	2,509	3.10 → 3.10	120.0 → 120.0	F → F	2,566
WB	2,337	3	2,339	2.88 → 2.89	120.0 → 120.0	F → F	2,392

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	2	50%
EB Through	1	25%
WB Right	1	25%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.14 Oakhurst Elementary School

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

charlotte-cdot-80097

NPA 240 · 0.97 mi from site

3

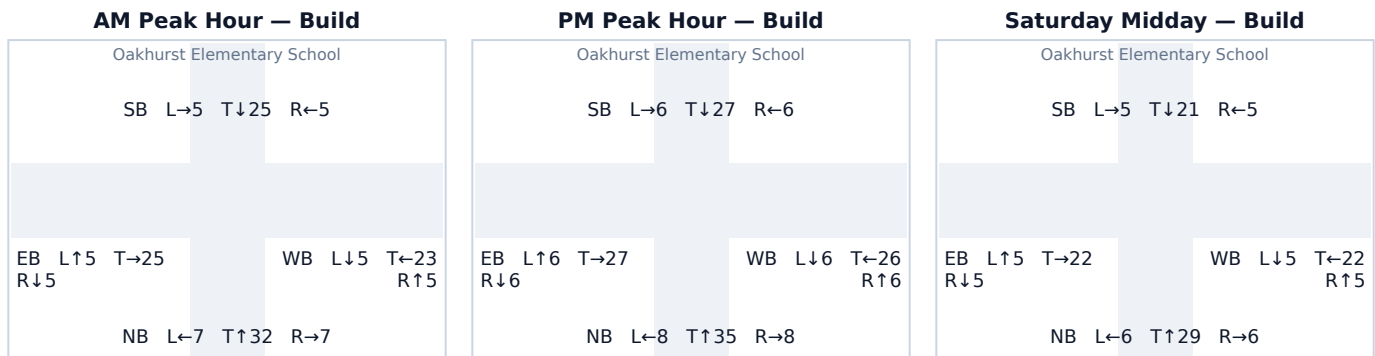
LOS B · 14.4 s/veh · v/c 0.09

LOS B · 14.4 s/veh · v/c 0.09

+0.0 s/veh

No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	51	0	51	0.06 → 0.06	14.2 → 14.2	B → B	30
SB	39	0	39	0.05 → 0.05	14.0 → 14.0	B → B	23
EB	38	1	39	0.05 → 0.05	14.0 → 14.0	B → B	23
WB	36	2	38	0.04 → 0.05	14.0 → 14.0	B → B	22

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	2	67%
EB Through	1	33%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.15 Albemarle Rd & Reddman Rd

Signal ID

Location

Added PM peak trips

Existing / No-Build (PM)

Build (PM)

Δ control delay

Mitigation

charlotte-172211825

NPA 164 · 0.98 mi from site

3

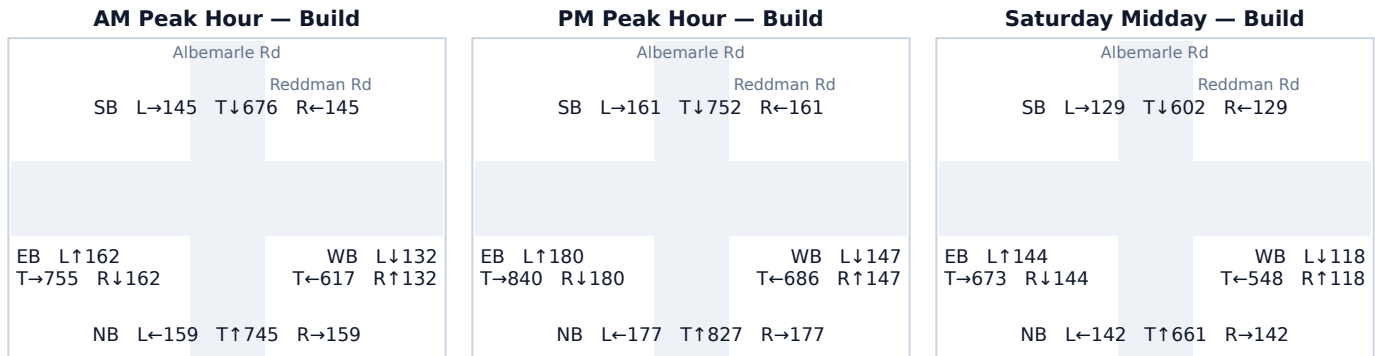
LOS F · 120.0 s/veh · v/c 2.46

LOS F · 120.0 s/veh · v/c 2.46

+0.0 s/veh

Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s.

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	1,181	0	1,181	1.46 → 1.46	120.0 → 120.0	F → F	1,208
SB	1,073	1	1,074	1.32 → 1.33	120.0 → 120.0	F → F	1,098
EB	1,198	2	1,200	1.48 → 1.48	120.0 → 120.0	F → F	1,227
WB	979	0	980	1.21 → 1.21	120.0 → 120.0	F → F	1,002

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	1	33%
SB Right	1	33%
EB Through	1	33%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.